

RQ2: Model and retrieval effects

Research question

RQ2. Does the choice of model or retrieval strategy affect what ends up in the profile?

The analysis compares final profile quality after deterministic validation, alongside extraction-token cost, latency, rejected findings, unresolved fields, and run-to-run variance.

Run configuration

- **60 completed profile builds:** 3 sites × 4 models × 3 retrieval modes, with additional repetitions for the stochastic expanded mode.
- **Sites:** Purdue Anvil, TACC Stampede3, and Notre Dame CRC.
- **Models:** GPT-5 Mini, GPT-5.6 Terra, Gemini Flash, and Gemini Pro.
- **Retrieval:** BM25, LLM-expanded BM25, and full corpus.
- **Repetitions:** one BM25 and one full-corpus run per site-model pair; three LLM-expanded BM25 runs per site-model pair.
- **Frozen inputs:** every run for a site uses the same login measurement, pilot result, and documentation corpus. No run performs new discovery, measurement, or pilot submission.
- **Balancing:** BM25 and full corpus have weight one; each expanded repetition has weight one third. Thus every site-model-retrieval configuration has equal aggregate weight.
- **Cost boundary:** extraction tokens and profile-build latency are compared here. One-time corpus-discovery tokens are excluded because the corpus is frozen and shared by every run at a site.

Results by model

Model	Field accuracy	Precision	Recall	Tokens	Latency (s)	Rejected	Unresolved
GPT-5 Mini	0.826	0.953	0.810	83,951	298.1	19.4	44.3
GPT-5.6 Terra	0.866	0.982	0.851	56,793	48.2	8.2	42.9
Gemini Flash	0.876	0.982	0.865	82,454	141.8	6.2	41.6
Gemini Pro	0.869	0.981	0.857	84,059	200.9	13.0	41.7

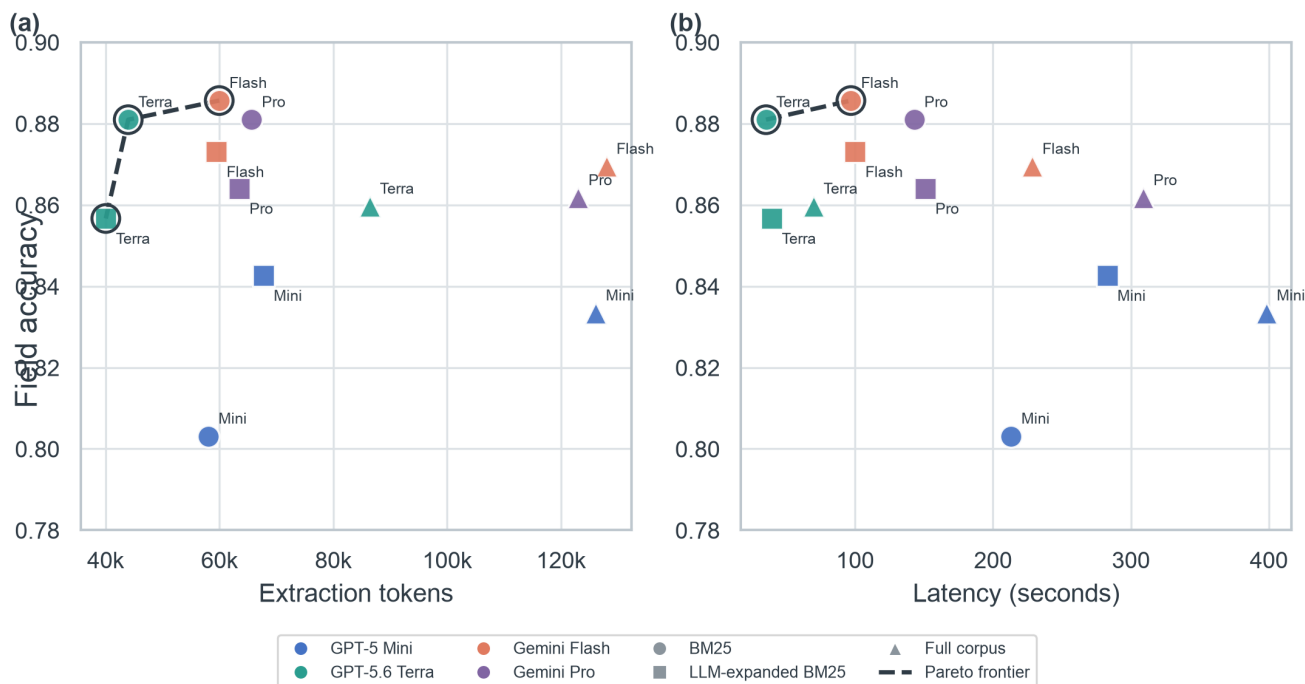
The model-level field-accuracy spread is 0.050 (5.0%). Gemini Flash has the highest balanced field accuracy. GPT-5.6 Terra is within 0.010 (1.0%) of that result while using 31.1% fewer extraction tokens and 66.0% less time than Gemini Flash.

Results by retrieval strategy

Retrieval	Field accuracy	Precision	Recall	Tokens	Latency (s)	Model calls	Rejected	Unresolved
BM25	0.863	0.972	0.849	56,900	122.1	4.7	5.0	42.1
LLM-expanded BM25	0.859	0.975	0.847	57,678	143.3	5.9	13.8	42.1
Full corpus	0.856	0.976	0.842	115,864	251.4	13.8	16.3	43.7

The retrieval-level field-accuracy spread is only 0.007 (0.7%). Relative to BM25, full corpus uses 2.04× as many extraction tokens and takes 2.06× as long without improving aggregate field accuracy. LLM-expanded BM25 uses 1.01× the tokens and 1.17× the time of BM25, also without an aggregate accuracy improvement.

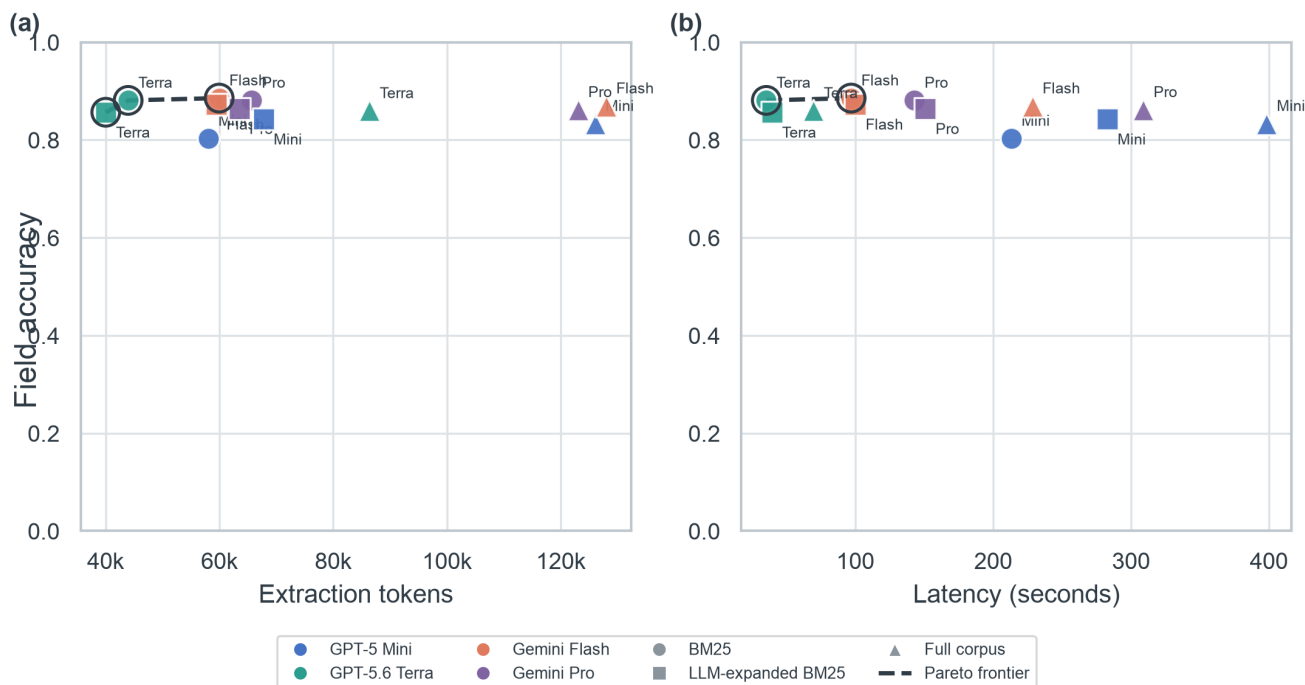
Quality-cost Pareto comparison



Vector PDF figure

Panel (a) compares field accuracy against extraction tokens; panel (b) compares accuracy against latency. Color identifies the model, point shape identifies the retrieval strategy, and all markers have the same size. The dashed line and outlined points mark the Pareto frontier: configurations for which no other point is both cheaper and at least as accurate. **The field-accuracy axis is truncated to expose differences among configurations.**

Full-scale Pareto reference



Vector PDF figure

This version uses the full 0–1 field-accuracy scale so the absolute magnitude of the differences remains visible.

Run-to-run variance

Across expanded-mode site-model groups, the mean field-accuracy standard deviation is 0.020. The largest is 0.075 for GPT-5 Mini on ND CRC. This variance is concentrated in particular site-model combinations rather than uniformly distributed.

Answer to RQ2

Retrieval strategy has little effect on aggregate profile quality in this experiment, but a large effect on cost: full corpus roughly doubles both tokens and latency. Model choice has a larger quality effect than retrieval choice, and an even larger latency effect. GPT-5.6 Terra provides the strongest quality-cost balance in this matrix: its accuracy is close to the best observed model while its token use and latency are substantially lower.

These are descriptive results over three sites. The deterministic validator constrains accepted outputs, but it does not make all models identical: individual hard fields and some site-model combinations still diverge.